



PRINT DATE: 2025-09-30 21:32:34 +0200

JOB ID: 1437b825-85b8-4ebb-868c-ab4437c4d4f3

RESULT PAGE: https://tygs.dsmz.de/user_results/show?guid=1437b825-85b8-4ebb-868c-ab4437c4d4f3

Table 1: Phylogenies

Publication-ready versions of both the genome-scale GBDP tree and the 16S rRNA gene sequence tree can be customized and exported either in SVG (vector graphic) or PNG format from within the phylogeny viewers in your TYGS result page. For publications the **SVG format is recommended** because it is lossless, always keeps its high resolution and can also be easily converted to other popular formats such as PDF or EPS. Please follow the link provided above!

Table 2: Identification

The below list contains the result of the TYGS species identification routine.

Explanation of remarks that might occur in the below table:

remark [R1]: The TYGS type strain database is automatically updated on an almost daily basis. However, if a particular type strain genome is not available in the TYGS database, this can have several reasons which are detailed in the FAQ. You can request an extended 16S rRNA gene analysis via the 16S tree viewer found in your result page to detect **not yet genome-sequenced** type strains relevant for your study.

remark [R2]: > 70% dDDH value (formula d_4) and (almost) minimal dDDH values for gene-content formulae d_0 and d_6 indicate a potentially unreliable identification result and should thus be checked via the 16S rRNA gene sequence similarity. Such strong deviations can, in principle, be caused by sequence contamination.

remark [R3]: G+C content difference of > 1 % indicates a potentially unreliable identification result because within species G+C content varies no more than 1 %, if computed from genome sequences (PMID: 24505073).

Strain	Conclusion	Identification result	Remark
'assembly.polished'	potential new species		see [R1]

Table 3: Pairwise comparisons of user genomes vs. type-strain genomes

The following table contains the pairwise dDDH values between your user genomes and the selected type-strain genomes. The dDDH values are provided along with their confidence intervals (C.I.) for the three different GBDP formulas:

- formula d_0 (a.k.a. GGDC formula 1): length of all HSPs divided by total genome length
- formula d_4 (a.k.a. GGDC formula 2): sum of all identities found in HSPs divided by overall HSP length
- formula d_6 (a.k.a. GGDC formula 3): sum of all identities found in HSPs divided by total genome length

Note: Formula d_4 is independent of genome length and is thus robust against the use of incomplete draft genomes. For other reasons for preferring formula d_4 , see the FAQ.

Query	Subject	d_0	C.I. d_0	d_4	C.I. d_4	d_6	C.I. d_6	Diff. G+C Percent
'assembly.polished.fasta'	<i>Flintibacter porci</i> P01025	14.9	[12.0 - 18.3]	26.4	[24.0 - 28.9]	15.1	[12.7 - 18.0]	3.07
'assembly.polished.fasta'	<i>Lawsonibacter hominis</i> NSJ-51	16.6	[13.6 - 20.1]	25.1	[22.8 - 27.6]	16.7	[14.1 - 19.7]	0.85
'assembly.polished.fasta'	<i>Pseudoflavonifractor capillosus</i> ATCC 29799	16.4	[13.5 - 19.9]	22.7	[20.4 - 25.1]	16.5	[13.9 - 19.4]	0.4
'assembly.polished.fasta'	<i>Pseudoflavonifractor gallinarum</i> DSM 107456	16.3	[13.3 - 19.8]	22.4	[20.1 - 24.8]	16.3	[13.7 - 19.2]	0.43
'assembly.polished.fasta'	<i>Intestinimonas aquisgranensis</i> CLA-AA-H199	16.6	[13.6 - 20.2]	21.8	[19.6 - 24.3]	16.6	[14.0 - 19.5]	0.32
'assembly.polished.fasta'	<i>Flavonifractor porci</i> P01024	16.1	[13.1 - 19.5]	21.1	[18.9 - 23.5]	16.1	[13.5 - 19.0]	1.08
'assembly.polished.fasta'	<i>Pseudoflavonifractor intestinhominis</i> CLA-AP-H29	16.4	[13.5 - 19.9]	21.0	[18.8 - 23.5]	16.4	[13.8 - 19.3]	2.03
'assembly.polished.fasta'	<i>Vermiculatibacterium agrestimuris</i> DSM 112226T	19.7	[16.6 - 23.3]	20.8	[18.6 - 23.2]	19.1	[16.4 - 22.1]	0.98
'assembly.polished.fasta'	<i>Intestinimonas timonensis</i> GD4 T	16.1	[13.1 - 19.6]	20.8	[18.5 - 23.2]	16.1	[13.5 - 19.0]	0.52
'assembly.polished.fasta'	<i>Clostridium orbiscindens</i> DSM 6740	15.4	[12.5 - 18.9]	20.7	[18.4 - 23.1]	15.5	[13.0 - 18.4]	0.84
'assembly.polished.fasta'	<i>Pseudoflavonifractor hominis</i> New-38	15.7	[12.8 - 19.2]	20.5	[18.3 - 23.0]	15.7	[13.2 - 18.6]	0.64
'assembly.polished.fasta'	<i>Candidatus Flavonifractor avistercoris</i> 6084	17.1	[14.1 - 20.6]	20.4	[18.2 - 22.8]	16.9	[14.3 - 19.9]	5.96
'assembly.polished.fasta'	<i>Intestinimonas butyriciproducens</i> DSM 26588	14.9	[12.0 - 18.3]	20.3	[18.1 - 22.8]	15.0	[12.6 - 17.9]	0.01
'assembly.polished.fasta'	<i>Lawsonibacter faecis</i> NSJ-52	15.2	[12.3 - 18.7]	20.3	[18.0 - 22.7]	15.3	[12.8 - 18.2]	2.48
'assembly.polished.fasta'	<i>Intestinimonas butyriciproducens</i> DSM 26588	14.9	[12.0 - 18.3]	20.2	[18.0 - 22.6]	15.0	[12.5 - 17.9]	0.09
'assembly.polished.fasta'	<i>Candidatus Enterenecus avicola</i> 153	14.5	[11.6 - 17.9]	19.1	[17.0 - 21.5]	14.6	[12.2 - 17.5]	1.4

Table 4: Strains in your dataset

Joint dataset of automatically determined closest type strains (if this mode was chosen), manually selected type strains (if selected accordingly) and the provided user strains, if provided (marked in **yellow**).

Strain	Authority	Other deposits	Synonyms	Base pairs	Percent G+C	No. proteins	Goldstamp	Bioproject accession	Biosample accession	Assembly accession	IMG OID
<i>Pseudoflavonifractor hominis</i> New-38	Liu et al. 2022	CGMCC 1.31611; KCTC 15862	<i>Pseudoflavonifractor hominis</i>	3050445	60.2	2870		PRJNA224116	SAMN15808717	GCF_014288125	
<i>Intestinimonas aquisgranensis</i> CLA-AA-H199	Afrizal et al. 2023	DSM 112680; JCM 35876	<i>Intestinimonas aquisgranensis</i>	3725314	59.8	3590		PRJNA767532	SAMN22081871	GCA_020687605	
<i>Lawsonibacter faecis</i> NSJ-52	Liu et al. 2022	CGMCC 1.32826; KCTC 25135	<i>Lawsonibacter faecis</i>	4048871	62.0	3436		PRJNA224116	SAMN15808702	GCF_014287675	
<i>Intestinimonas butyriciproducens</i> DSM 26588	Kläring et al. 2013	CCUG 63529; SRB-521-5-I	<i>Intestinimonas butyriciproducens</i>	3224506	59.5	3134	Gp0444085	PRJNA224116	SAMN03610663	GCF_004154955	
<i>Candidatus Flavonifractor avistercoris</i> 6084	Gilroy et al. 2021		<i>Candidatus Flavonifractor avistercoris</i>	2617301	65.5	2499		PRJNA543206	SAMN15816821	GCA_019119085	
<i>Candidatus Enterenecus avicola</i> 153	Gilroy et al. 2021		<i>Candidatus Enterenecus avicola</i>	1738458	60.9	1577		PRJNA543206	SAMN15817108	GCA_018715985	
<i>Lawsonibacter hominis</i> NSJ-51	Liu et al. 2022	CGMCC 1.32825; KCTC 25134	<i>Lawsonibacter hominis</i>	3476845	60.4	3256		PRJNA224116	SAMN15808689	GCF_014287895	
<i>Clostridium orbiscindens</i> DSM 6740	Winter et al. 1991	265; ATCC 49531	<i>Clostridium orbiscindens</i>	4428335	60.4	4282	Gp0325695	PRJNA519318	SAMN10864851	GCA_004345805	2795385455
<i>Intestinimonas butyriciproducens</i> DSM 26588	Kläring et al. 2013	CCUG 63529; SRB-521-5-I	<i>Intestinimonas butyriciproducens</i>	3199160	59.4	3122	Gp0251946	PRJNA439840	SAMN08769261	GCA_003096335	2756170247

Strain	Authority	Other deposits	Synonyms	Base pairs	Percent G+C	No. proteins	Goldstamp	Bioproject accession	Biosample accession	Assembly accession	IMG OID
<i>Pseudoflavonifactor capillosus</i> ATCC 29799	(Tissier 1908) Carlier et al. 2010	CCUG 15402 A; CCUG 15402; DSM 23940; VPI R2-29-1	<i>Bacillus capillosus</i> ; <i>Bacteroides capillosus</i> ; <i>Pseudoflavonifactor capillosus</i>	4241 076	59.1	4829	Gp0000895	PRJNA18173	SAMN00627059	GCA_000169255	640963014
<i>Intestinimonas timonensis</i> GD4 T	Durand et al. 2017	CSUR P2010	<i>Intestinimonas timonensis</i>	3441 933	59.0	3481		PRJNA224116	SAMEA5138309	GCF_900626145	
<i>Vermiculatibacterium agrestimuris</i> DSM 112226T	Afrizal et al. 2022	CLA AA-M16	<i>Vermiculatibacterium agrestimuris</i>	2231 797	60.5	1984		PRJEB50452	SAMEA14418538	GCA_943193085	
<i>Flintibacter porci</i> P01025	Niu et al. 2025	CGMCC 1.18060; KCTC 25794	<i>Flintibacter porci</i>	3355 411	56.5	3214		PRJNA224116	SAMN43505365	GCF_041925195	
<i>Flavonifactor porci</i> P01024	Niu et al. 2025	CGMCC 1.18055; KCTC 25793	<i>Flavonifactor porci</i>	3344 173	58.4	3150		PRJNA224116	SAMN40517927	GCF_037572415	
<i>Pseudoflavonifactor intestinihominis</i> CLA-AP-H29	Hitch et al. 2025	DSM 118073; CLA-AP-H29T (phylum Bacillota, family Oscillospiraceae); LMG 33601	<i>Pseudoflavonifactor intestinihominis</i>	3687 158	61.5	3309		PRJNA996881	SAMN40466817	GCA_040096055	
<i>Pseudoflavonifactor gallinarum</i> DSM 107456	Zenner et al. 2021	JCM 34594; Cla-CZ-98	<i>Pseudoflavonifactor gallinarum</i>	3177 256	59.9	3000		PRJNA668258	SAMN16401881	GCA_014982855	
assembly.polished.fasta				2617 187	59.5	2593					

Methods, Results and References

The genome sequence data were uploaded to the Type (Strain) Genome Server (TYGS), a free bioinformatics platform available under <https://tygs.dsmz.de>, for a whole genome-based taxonomic analysis [1]. The analysis also made use of recently introduced methodological updates and features [2]. Information on nomenclature, synonymy and associated taxonomic literature was provided by TYGS's sister database, the List of Prokaryotic names with Standing in Nomenclature (LPSN, available at <https://lpsn.dsmz.de>) [2]. The results were provided by the TYGS on 2025-09-30. The TYGS analysis was subdivided into the following steps:

Determination of closely related type strains

Determination of closest type strain genomes was done in two complementary ways: First, all user genomes were compared against all type strain genomes available in the TYGS database via the MASH algorithm, a fast approximation of intergenomic relatedness [3], and, the ten type strains with the smallest MASH distances chosen per user genome. Second, an additional set of ten closely related type strains was determined via the 16S rDNA gene sequences. These were extracted from the user genomes using RNAmmer [4] and each sequence was subsequently BLASTed [5] against the 16S rDNA gene sequence of each of the currently 23760 type strains available in the TYGS database. This was used as a proxy to find the best 50 matching type strains (according to the bitscore) for each user genome and to subsequently calculate precise distances using the Genome BLAST Distance Phylogeny approach (GBDP) under the algorithm 'coverage' and distance formula d_5 [6]. These distances were finally used to determine the 10 closest type strain genomes for each of the user genomes.

Pairwise comparison of genome sequences

For the phylogenomic inference, all pairwise comparisons among the set of genomes were conducted using GBDP and accurate intergenomic distances inferred under the algorithm 'trimming' and distance formula d_5 [6]. 100 distance replicates were calculated each. Digital DDH values and confidence intervals were calculated using the recommended settings of the GGDC 4.0 [2,6].

Phylogenetic inference

The resulting intergenomic distances were used to infer a balanced minimum evolution tree with branch support via FASTME 2.1.6.1 including SPR postprocessing [7]. Branch support was inferred from 100 pseudo-bootstrap replicates each. The trees were rooted at the midpoint [8] and visualized with PhyD3 [9].

Type-based species and subspecies clustering

The type-based species clustering using a 70% dDDH radius around each of the 16 type strains was done as previously described [1]. The resulting groups are shown in Table 1 and 4. Subspecies clustering was done using a 79% dDDH threshold as previously introduced [10].

Results

Type-based species and subspecies clustering

The resulting species and subspecies clusters are listed in Table 4, whereas the taxonomic identification of the query strains is found in Table 1. Briefly, the clustering yielded 15 species clusters and the provided query strains were assigned to 1 of these. Moreover, user strains were located in 1 of 15 subspecies clusters.

Figure caption SSU tree

Figure 1. Tree inferred with FastME 2.1.6.1 [7] from GBDP distances calculated from 16S rDNA gene sequences. The branch lengths are scaled in terms of GBDP distance formula d_5 . The numbers above branches are GBDP pseudo-bootstrap support values > 60 % from 100 replications, with an average branch support of 88.2 %. The tree was rooted at the midpoint [8].

Figure caption genome tree

Figure 2. Tree inferred with FastME 2.1.6.1 [7] from GBDP distances calculated from genome sequences. The branch lengths are scaled in terms of GBDP distance formula d_5 . The numbers above branches are GBDP pseudo-bootstrap support values > 60 % from 100 replications, with an average branch support of 65.9 %. The tree was rooted at the midpoint [8].

References

- [1] Meier-Kolthoff JP, Göker M. TYGS is an automated high-throughput platform for state-of-the-art genome-based taxonomy. *Nat. Commun.* 2019;10: 2182. DOI: 10.1038/s41467-019-10210-3
- [2] Meier-Kolthoff JP, Sardà Carbasse J, Peinado-Olarte RL, Göker M. TYGS and LPSN: a database tandem for fast and reliable genome-based classification and nomenclature of prokaryotes. *Nucleic Acid Res.* 2022;50: D801–D807. DOI: 10.1093/nar/gkab902
- [3] Ondov BD, Treangen TJ, Melsted P, et al. Mash: Fast genome and metagenome distance estimation using MinHash. *Genome Biol* 2016;17: 1–14. DOI: 10.1186/s13059-016-0997-x
- [4] Lagesen K, Hallin P. RNAmmer: consistent and rapid annotation of ribosomal RNA genes. *Nucleic Acids Res.* Oxford Univ Press; 2007;35: 3100–3108. DOI: 10.1093/nar/gkm160
- [5] Camacho C, Coulouris G, Avagyan V, Ma N, Papadopoulos J, Bealer K, et al. BLAST+: architecture and applications. *BMC Bioinformatics.* 2009;10: 421. DOI: 10.1186/1471-2105-10-421
- [6] Meier-Kolthoff JP, Auch AF, Klenk H-P, Göker M. Genome sequence-based species delimitation with confidence intervals and improved distance functions. *BMC Bioinformatics.* 2013;14: 60. DOI: 10.1186/1471-2105-14-60
- [7] Lefort V, Desper R, Gascuel O. FastME 2.0: A comprehensive, accurate, and fast distance-based phylogeny inference program. *Mol Biol Evol.* 2015;32: 2798–2800. DOI: 10.1093/molbev/msv150
- [8] Farris JS. Estimating phylogenetic trees from distance matrices. *Am Nat.* 1972;106: 645–667.
- [9] Kreft L, Botzki A, Coppens F, Vandepoele K, Van Bel M. PhyD3: A phylogenetic tree viewer with extended phyloXML support for functional genomics data visualization. *Bioinformatics.* 2017;33: 2946–2947. DOI: 10.1093/bioinformatics/btx324
- [10] Meier-Kolthoff JP, Hahnke RL, Petersen J, Scheuner C, Michael V, Fiebig A, et al. Complete genome sequence of DSM 30083^T, the type strain (U5/41^T) of *Escherichia coli*, and a proposal for delineating subspecies in microbial taxonomy. *Stand Genomic Sci.* 2014;9: 2. DOI: 10.1186/1944-3277-9-2